CS246: Database Management Systems Lab

Week # 02 (1 Questions, 100 Marks)
Lab session: AL1
Held on: 19-Jan-2026 (Mon)
Lab Timings: 14:00 to 17:00 Hours Pages: 2
Submission dead line: 17:00 hrs, 19-Jan-2026
Submissions accepted from: 16:45 hrs, 19-Jan-2026
Submissions accepted till: 17:30 hrs, 19-Jan-2026
Every one minute delay in submission will result in one mark deduction
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Question 1: (100 points)

Given the user current location in terms of `(latitude, longitude)` obtain nearest ATMs within 1 kilometer range of the specified bank. Follow the detailed problem description as given below:

Task 01 (4 marks) Declare a structure ATM whose data members are

1. A string to hold the bank name
2. A `double` to store `latitude`
3. A `double` to store `longitude`
4. A string to hold the ATM ID

Task 02 (2 marks) Read the data from the input file `ATMs.csv` whose format is as described

1. 1^{st} column contains Bank name
2. 2^{nd} column contains latitude information
3. 3^{rd} column contains longitude information
4. 4^{th} column contains ATM ID

Task 03 (2 marks) Declare a query location of type ATM structure

Task 04 (2 marks) Read query locations from the file `query.csv` whose file format is identical to the input file formation described in Task 02. The file contain five lines, each in the same format as `ATMs.csv`:

```
bank_name,latitude,longitude,user-id
```

For each line (each query location), treat the latitude and longitude as a separate user position. The bank name in the query file indicates which bank's ATMs to search for in this query. The user_id field can be ignored.

Task 05 (50 marks) For each query location in `query.csv`, find all ATMs of the specified bank that are within 1 km of that query location. Sort them in increasing order of distance (implement Bubble sort algorithm – library sort functions are not allowed).

Use the formula given below to compute approximate distance in kilometers given between (`lat`$_1$, `long`$_1$) and (`lat`$_2$, `long`$_2$)

$$dist_{km} = 111.0 \times \sqrt{(lat_2 - lat_1)^2 + (long_2 - long_1)^2}$$

1. You must implement bubble sort algorithm
2. Do not use `std::sort, qsort`, or any library sorting function.
3. Using in-built library function results in 0 marks.
4. If two or more ATMs have the same distance, sort them in ascending alphabetical order of their ATM number (string comparison).

Task 06 (40 marks) (Each query bank 8 marks) Output:

1. Print distances rounded to one decimal place using fixed notation (example: 0.9 km, 1.0 km).
2. Use dist $\leq$ 1.000001 when checking the 1 km range to avoid floating-point precision issues.
3. If more than one ATM location is found within 1 kilometer range, you should provide all such ATMs.
4. Example output is

```
Nearby SBI ATMs within 1 km of location (17.6955, 83.2328):

SBI-ATM-38 dist: 0.9 km
SBI-ATM-39 dist: 0.9 km
SBI-ATM-37 dist: 0.9 km
SBI-ATM-40 dist: 0.9 km
SBI-ATM-36 dist: 0.9 km
SBI-ATM-41 dist: 1.0 km
SBI-ATM-35 dist: 1.0 km
```

5. If no ATM is found within 1 kilometer range you should output `NO ATM is found within 1 kilometer range`